

The Hemispherical Higgs Oscillator: Exact Actions, Friedrichs Spectrum, and Identity Revivals

HCS-C373 theorem package

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Abstract

We close the classical and Friedrichs quantum mechanics of the isotropic Higgs oscillator on one open hemisphere, while keeping its singular boundary explicit. For positive classical coupling, a turning-point integral gives the exact radial action $I_r = (J - |L| - \omega R^2)/2$, where $J = (2R^2 E + \omega^2 R^4)^{1/2}$. The Hamiltonian in actions has an exact 2:1 frequency lock, so every regular trajectory closes; circular, meridional, equilibrium, and zero-coupling faces are separated. For the quantum Friedrichs realization with nonnegative coupling, Jacobi modes have $E_N = \hbar^2(N+1)(N+1+2\nu)/(2R^2)$ and multiplicity $N+1$. Consecutive spectral gaps then prove that the propagator is the identity precisely when $\tau = \pi M$ and $M(3+2\nu)$ is even. Thus a positive identity revival exists if and only if 2ν is rational, with an explicit least time. Exact finite enumeration is supplied only as a reproducibility receipt.

1 Positive-coupling classical system

Fix $R > 0, \omega > 0$, and polar coordinates $0 \leq \theta < \pi/2, \phi \in \mathbb{R}/(2\pi\mathbb{Z})$. With $L = p_\phi$, consider

$$H = \frac{p_\theta^2}{2R^2} + \frac{L^2}{2R^2 \sin^2 \theta} + \frac{\omega^2 R^2}{2} \tan^2 \theta. \quad (1)$$

The positive coupling is part of the theorem: its divergent equatorial barrier confines every finite-energy trajectory away from the missing equator. Put

$$J = (2R^2 E + \omega^2 R^4)^{1/2}, \quad x = \sin^2 \theta.$$

Theorem 1 (Turning points, action, and inversion). *For a regular trajectory with $L \neq 0$, the two turning points $0 < x_- < x_+ < 1$ are the roots of*

$$J^2 x^2 - (2R^2 E + L^2)x + L^2 = 0. \quad (2)$$

The radial action and the Hamiltonian in actions are

$$I_r = \frac{1}{2\pi} \oint p_\theta d\theta = \frac{J - |L| - \omega R^2}{2}, \quad (3)$$

$$H(I_r, L) = \frac{(2I_r + |L| + \omega R^2)^2 - \omega^2 R^4}{2R^2}. \quad (4)$$

The allowed chamber is $E \geq \omega|L| + L^2/(2R^2)$, with equality exactly on the circular face.

Proof. At energy E , direct rearrangement of (1) gives

$$p_\theta^2 = 2R^2 E - \frac{L^2}{x} - \frac{\omega^2 R^4 x}{1-x} = \frac{J^2(x-x_-)(x_+-x)}{x(1-x)}.$$

Because $d\theta = dx/(2\sqrt{x(1-x)})$, the two branches of the closed radial loop give

$$I_r = \frac{J}{2\pi} \int_{x_-}^{x_+} \frac{\sqrt{(x-x_-)(x_+-x)}}{x(1-x)} dx. \quad (5)$$

For $0 < a < b < 1$, substitute $x = a + (b-a)\sin^2 u$ and split $1/[x(1-x)] = 1/x + 1/(1-x)$. The two elementary integrals yield

$$\int_a^b \frac{\sqrt{(x-a)(b-x)}}{x(1-x)} dx = \pi \{1 - \sqrt{ab} - \sqrt{(1-a)(1-b)}\}. \quad (6)$$

Vieta's relations for (2) are

$$x_- x_+ = \frac{L^2}{J^2}, \quad (1 - x_-)(1 - x_+) = \frac{\omega^2 R^4}{J^2}.$$

Equations (5) and (6) prove (3); solving for J proves (4). The two roots coalesce exactly at the displayed energy threshold. $\square$

Theorem 2 (Frequency lock and periods). *On either signed chamber $L > 0$ or $L < 0$,*

$$\Omega_r = \frac{\partial H}{\partial I_r} = \frac{2J}{R^2}, \quad \Omega_\phi = \frac{\partial H}{\partial L} = \text{sgn}(L) \frac{J}{R^2}.$$

Every regular trajectory has radial period $\pi R^2/J$ and primitive phase-space period $2\pi R^2/J$.

Proof. Differentiate (4). During one radial return the azimuthal angle advances by $\pi \text{sgn}(L)$, so it does not return to the same phase-space point. Two radial returns give angular increments 4π and $2\pi \text{sgn}(L)$ in the two action angles, proving primitivity. $\square$

2 Classical boundary atlas

The action formula has four distinct boundary interpretations.

face	exact behavior
circular	$I_r = 0, L \neq 0$; the double root is $\sin^2 \theta_c = L /(L + \omega R^2)$ and the angular period is $2\pi R^2/J$.
meridional	$L = 0, I_r > 0$; the roots are 0 and $1 - (\omega R^2/J)^2$. A Cartesian tangent chart continues the orbit through the polar-coordinate singularity.
north equilibrium	$I_r = L = 0$, hence $E = 0$ and the orbit is fixed.
zero coupling	$\omega = 0$ removes the equatorial barrier. The open hemisphere is then incomplete for crossing geodesics, so it is excluded from the classical periodic-flow theorem.

This separation prevents a formula valid on regular Liouville tori from being silently extended through collapsed or incomplete faces.

3 Friedrichs quantization on one hemisphere

Now allow $\omega \geq 0$, retain $R, \hbar > 0$, and take the Friedrichs realization of

$$\hat{H} = -\frac{\hbar^2}{2R^2} \left[\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2 \right] + \frac{\omega^2 R^2}{2} \tan^2 \theta. \quad (7)$$

Define

$$\nu = \sqrt{(\omega R^2/\hbar)^2 + \frac{1}{4}}.$$

Theorem 3 (Jacobi spectrum). *For $m \in \mathbb{Z}$, $n_r \in \mathbb{Z}_{\geq 0}$, and $N = 2n_r + |m|$, a separated eigenfunction, up to normalization, is*

$$\psi_{n_r, m} = e^{im\phi} (\sin \theta)^{|m|} (\cos \theta)^{\nu+1/2} P_{n_r}^{(|m|, \nu)}(\cos 2\theta), \quad (8)$$

with

$$E_N = \frac{\hbar^2}{2R^2} (N+1)(N+1+2\nu). \quad (9)$$

The multiplicity of E_N on one hemisphere is $N+1$, and these modes form a complete orthogonal basis of the Friedrichs realization.

Proof. Separation with $e^{im\phi}$ gives a singular Sturm–Liouville problem. After extracting the north-pole factor $(\sin \theta)^{|m|}$ and the Friedrichs equatorial factor $(\cos \theta)^{\nu+1/2}$, set $y = \cos 2\theta$. The remaining factor v satisfies

$$(1 - y^2)v'' + [\nu - |m| - (|m| + \nu + 2)y]v' + n_r(n_r + |m| + \nu + 1)v = 0, \quad (10)$$

the Jacobi equation. Polynomial termination gives (8) and (9). Jacobi completeness with its positive weight, followed by Fourier completeness in ϕ , proves completeness.

For fixed N , admissible labels obey $|m| \leq N$ and $N - |m| \in 2\mathbb{Z}$. Directly counting the two signs for $m \neq 0$ and one copy for $m = 0$ gives $N+1$. This count is the proof of multiplicity; we do not use a conflicting prose degeneracy sentence in the historical separation source. $\square$

Limits and domain. For fixed $N, \omega, \hbar$ and $R \rightarrow \infty$, $\nu = \omega R^2/\hbar + o(R^2)$, so $E_N \rightarrow \hbar\omega(N+1)$. At $\omega = 0$, $\nu = 1/2$ and

$$E_N = \frac{\hbar^2}{2R^2}(N+1)(N+2).$$

Writing $l = N + 1$, these are the Dirichlet-hemisphere levels $\hbar^2 l(l+1)/(2R^2)$ with multiplicity l . They are the equator-odd sector, not the full-sphere spectrum with multiplicity $2l+1$. Thus $\omega = 0$ is valid quantum mechanically only because the Friedrichs domain supplies the Dirichlet boundary omitted by the classical open flow.

References

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